AI-Based Suspicious Mail Detector

1. Introduction

Phishing emails are one of the most common cyber security threats. Attackers often try to trick users by pretending to be trusted organizations, creating a sense of urgency, or asking for sensitive information such as passwords, account details, or payment information.

The aim of this project is to develop a simple web-based suspicious mail detection demo. The application allows users to enter the sender name, sender email address, and email body. Then, the system analyzes the given information and generates a risk report for the user.

This demo version does not use a real artificial intelligence API. Instead, it simulates an AI-based analysis approach by checking sender identity, suspicious phrases, unsafe links, and social engineering patterns. In a real-world version, this analysis engine can be replaced with a pre-trained artificial intelligence model or an LLM API.

2. Project Purpose

The main purpose of this project is to help users understand whether an email may be suspicious or dangerous before interacting with it.

The application focuses on three main inputs:

- Sender name
- Sender email address
- Email body

Based on these inputs, the system produces:

- Risk level
- Risk score
- Sender analysis
- Content analysis
- Suspicious phrases
- Suspicious links
- Security feedback

This makes the project not only a simple detector, but also an explainable security tool.

3. System Overview

The application is designed as a frontend-only web demo. It can run directly in a browser without any backend server or database.

The project consists of three main files:

```
AI-Based-Suspicious-Mail-Detector/
│
├── index.html
├── style.css
├── script.js
└── README.md
```

The index.html file contains the structure of the web page.
The style.css file controls the visual design.
The script.js file contains the suspicious mail analysis logic.
The README.md file includes test email examples for Safe, Medium Risk, and High Risk levels

4. User Interface

The user interface is designed to be clean and simple. It includes input fields where the user can enter email information.

The main form contains:

- Sender Name input
- Sender Email Address input
- Email Body text area
- Analyze Email button
- Clear button

After the user clicks the analyze button, the result section displays the generated risk report.

Screenshot 1: Main Page Interface

5. Application Workflow

The application follows a simple workflow:

User enters email information
↓
User clicks Analyze Email
↓
JavaScript analysis engine checks the input
↓
Risk score is calculated
↓
Risk level is determined
↓
Detailed report is displayed to the user

The application does not automatically open or execute any link inside the email. All email content is treated as untrusted text.

6. Analysis Method

The demo uses an AI-like scoring engine. This means that the system behaves like an AI-assisted detector, but the current version works locally with JavaScript-based logic.

The system analyzes the following elements:

6.1 Sender Analysis

The sender name and sender email address are checked together. For example, if the sender name claims to be a well-known company but the email address uses a free provider such as Gmail, Outlook, or Yahoo, the system marks this as suspicious.

Example:

Sender Name: PayPal Support
Sender Email: paypal.security@gmail.com

This is suspicious because a trusted company should normally use its official domain.

6.2 Content Analysis

The email body is analyzed for common phishing indicators such as:

- Urgent language
- Account verification requests
- Password requests
- Financial terms
- Threatening language
- Suspicious instructions
- Generic greetings

6.3 Link Analysis

The system also checks links inside the email body. Shortened links and HTTP links are considered more suspicious.

Examples of suspicious links:

http://bit.ly/secure-login
http://tinyurl.com/account-update

Shortened links can hide the real destination, so they increase the risk score.

7. Risk Scoring

The system calculates a risk score between 0 and 100. Different suspicious indicators add points to the score.

Example scoring logic:

Suspicious sender identity: +25
Urgent language: +15
Password request: +25
Account suspension threat: +20
Shortened link: +25
HTTP link: +10
Generic greeting: +5

The final score is limited to 100.

Risk levels are classified as follows:

Risk Score	Risk Level
0–30	Safe
31–65	Medium Risk
66–100	High Risk

8. Test Cases

The project includes test email examples in the README.md file. There are two examples for each risk level.

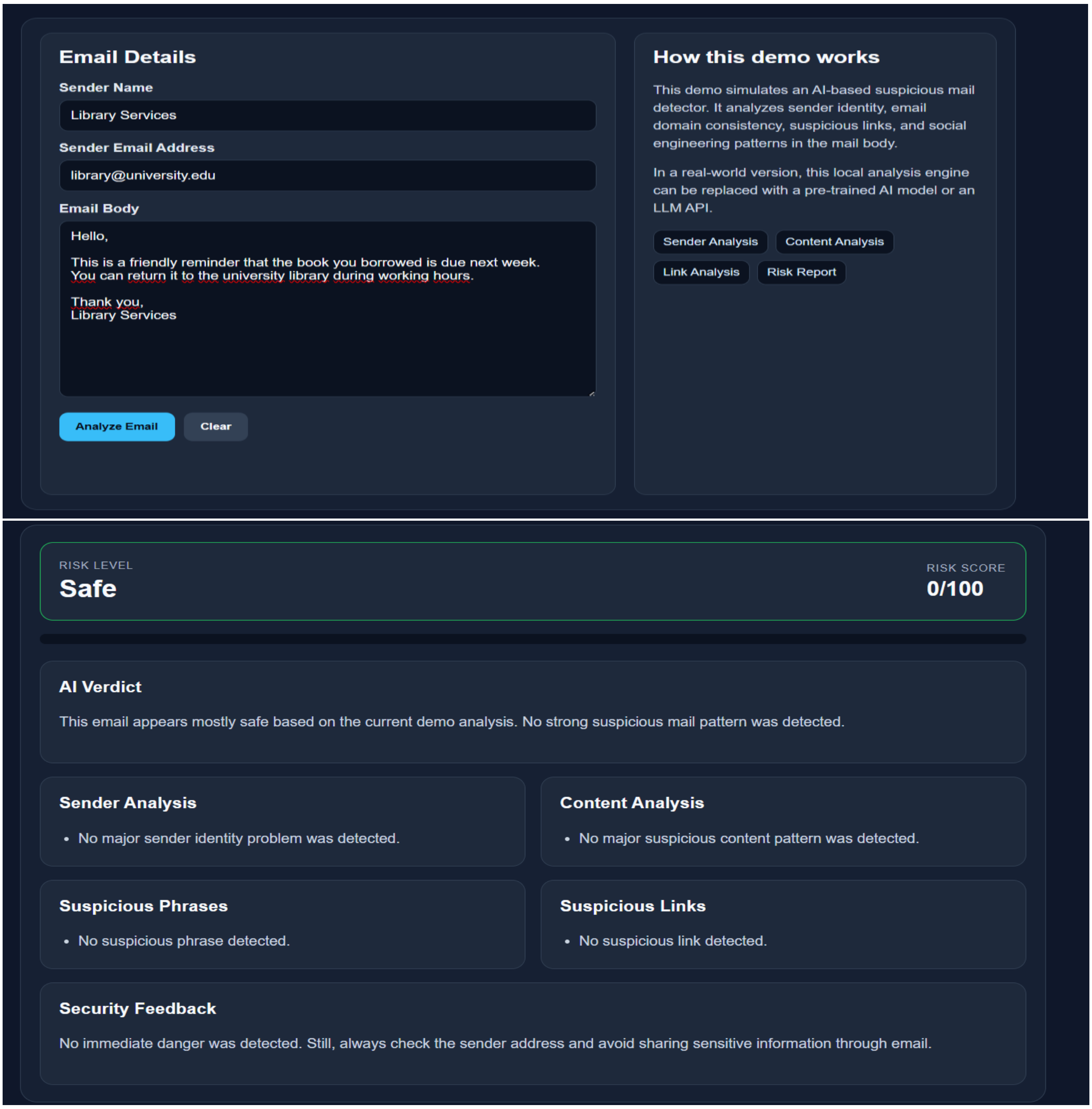
8.1 Safe Email Test

A safe email usually does not contain suspicious links, password requests, threats, or urgent instructions.

Expected Result:

Risk Level: Safe
Risk Score: Low

Screenshot 2: Safe Email Result



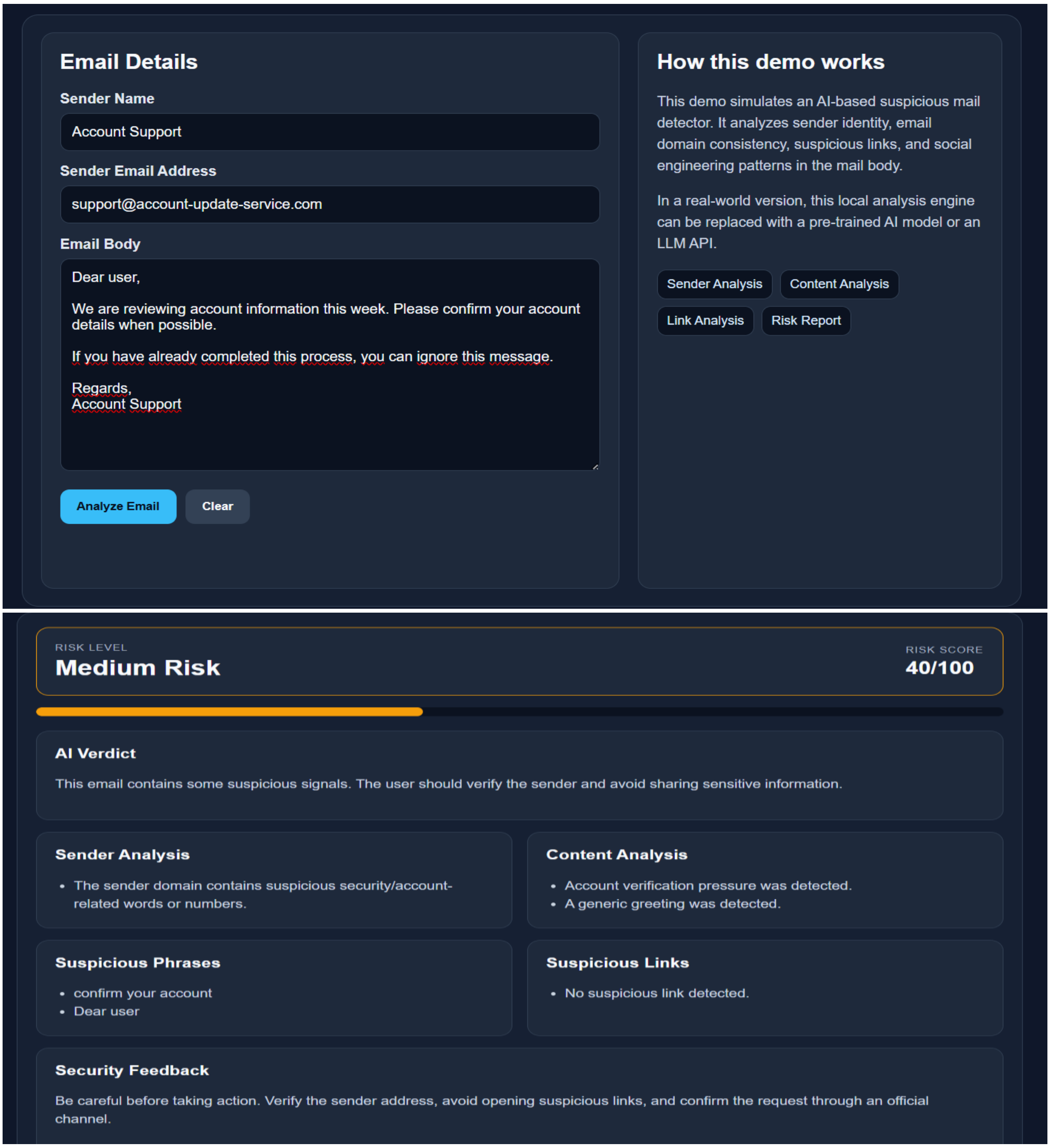
8.2 Medium Risk Email Test

A medium-risk email may contain some suspicious elements, such as payment-related language or a generic sender address, but it may not directly ask for passwords or contain clearly malicious links.

Expected Result:

Risk Level: Medium Risk
Risk Score: Medium

Screenshot 3: Medium Risk Email Result



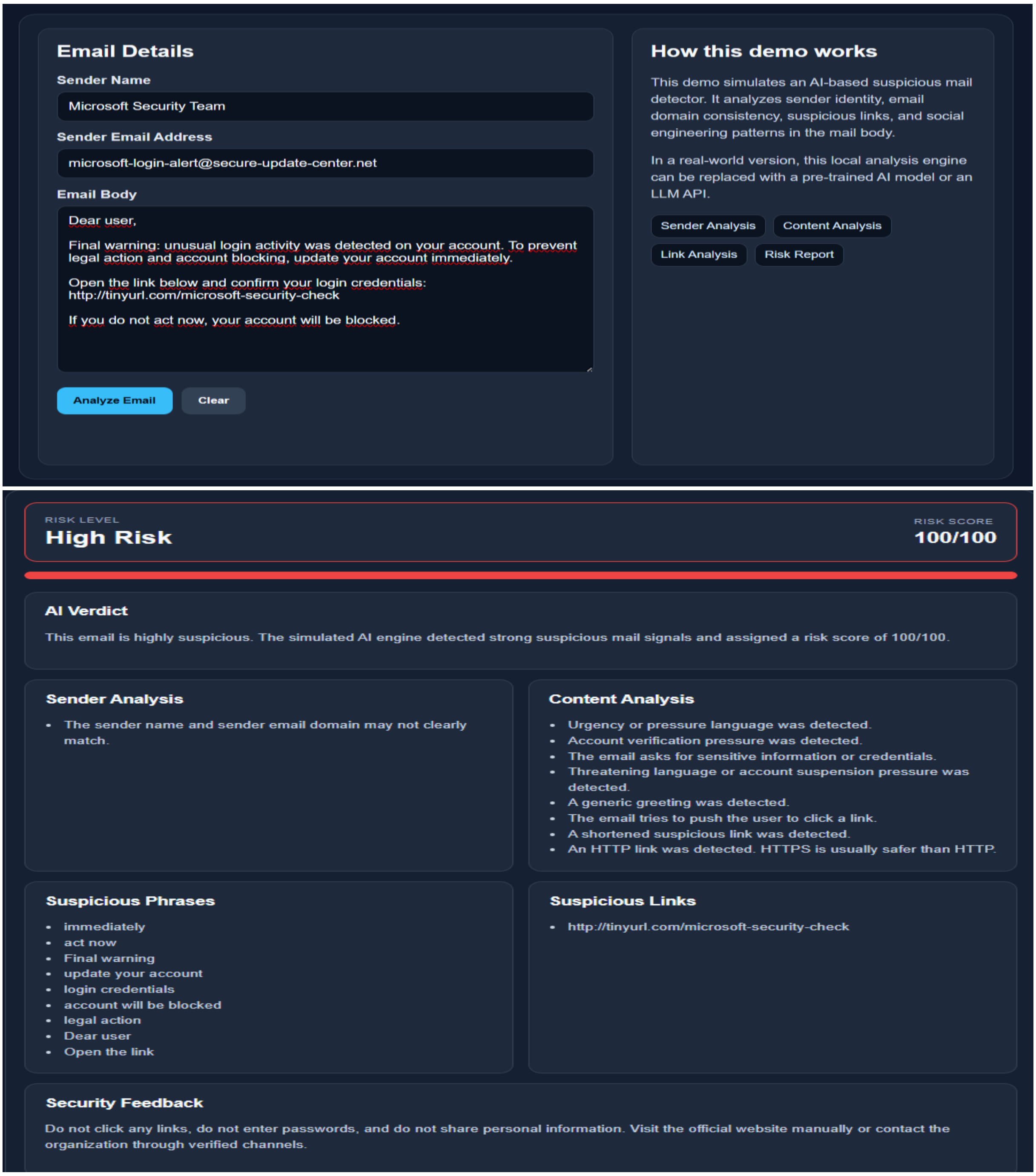
8.3 High Risk Email Test

A high-risk email usually contains multiple phishing indicators. For example, it may pretend to be a trusted company, use urgent language, threaten account suspension, ask for a password, and include a shortened suspicious link.

Expected Result:

Risk Level: High Risk
Risk Score: High

Screenshot 4: High Risk Email Result



9. Result Report

After analysis, the application generates a detailed report. The report includes several sections:

Risk Level

This shows whether the email is Safe, Medium Risk, or High Risk.

Risk Score

This shows the calculated score between 0 and 100.

Sender Analysis

This explains whether the sender identity looks suspicious.

Content Analysis

This explains suspicious patterns found in the email body.

Suspicious Phrases

This section lists risky phrases detected in the email.

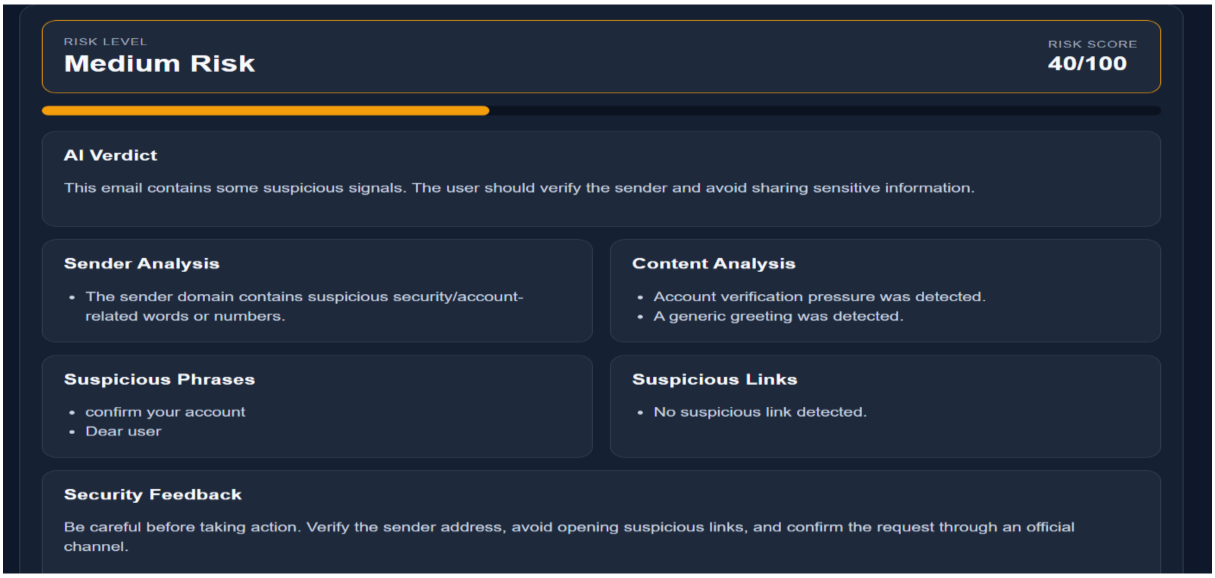
Suspicious Links

This section lists suspicious links found in the email.

Security Feedback

This section gives advice to the user, such as not clicking suspicious links or not sharing passwords.

Screenshot 5: Detailed Risk Report



10. Technologies Used

The project uses basic web technologies:

Technology	Purpose
HTML	Page structure
CSS	Page design and layout
JavaScript	Email analysis and dynamic result generation
README.md	Test email documentation

The project does not require:

- Backend server
- Database
- API key
- Internet connection
- Installation

This makes the demo easy to run during a presentation.

11. Limitations

This project is a demo version. It does not use a real trained machine learning model or live AI API.

Because of this, there are some limitations:

- It may not detect every phishing email.
- It may classify some emails incorrectly.
- It cannot verify real sender reputation.
- It cannot check live domain records.
- It cannot analyze attachments.
- It cannot scan real URLs in real time.

However, the demo successfully shows the basic idea of an AI-assisted suspicious mail detector.

12. Future Improvements

In the future, this project can be improved by adding a real AI model or API.

Possible improvements include:

- Integration with a pre-trained phishing detection model
- LLM API integration
- Real-time URL reputation checking
- Email header analysis
- Attachment scanning
- Domain reputation analysis
- Browser extension version
- User history-based risk detection

With these improvements, the system could become more accurate and closer to a real-world cyber security solution.

13. Conclusion

The AI-Based Suspicious Mail Detector is a simple web-based cyber security demo that helps users identify suspicious emails. The application analyzes sender information, email content, suspicious phrases, and unsafe links. Then, it generates an explainable risk report with a risk level, risk score, and security feedback.

Although this demo does not use a real AI model, it simulates the behavior of an AI-assisted phishing detection system. It is suitable for demonstrating how suspicious email detection can work in a user-friendly and understandable way.

The project can be further developed by replacing the local analysis engine with a real pre-trained artificial intelligence model or LLM API.